Motors and generators

antocciphysics | TEACHER KEY | CPA

Learning target: Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 12; 3 course days later.*
A stronger field at A in that drawing convention; field lines are a model, not physical strands.

02. Explain a worked example (worked)
Motor electrical to mechanical; generator mechanical to electrical; both may have thermal losses.

03. Look and explain (model)
Electrical to mechanical for motor; mechanical to electrical for generator.

04. Build the solution (practice)
60%; 40 J other transfers.

05. Use the idea (reason)
Each conversion transfers some energy to surroundings; usable energy in the loop decreases.

06. Choose a method (mixed) *Review: Unit 7, Day 3; 12 course days later.*
Positive charge: left. Negative charge: right. Each force has magnitude 24 N.

07. Exit check (exit)
Generator.

08. STRETCH: Apply it another way (stretch)
Higher useful mechanical power at fixed input means greater efficiency; measure useful work/time and electrical V, I.

Motors and generators

antocciphysics | TEACHER KEY | HONORS

Learning target: Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 12; 3 course days later.*
Each piece is normally a smaller magnet with both poles; cutting does not isolate a magnetic monopole.

02. Explain a worked example (worked)
Motor electrical to mechanical; generator mechanical to electrical; both may have thermal losses.

03. Represent and interpret (model)
Electrical to mechanical for motor; mechanical to electrical for generator.

04. Solve and justify (practice)
36 W; 72%; 14 W.

05. Analyze the reasoning (reason)
Each conversion transfers some energy to surroundings; usable energy in the loop decreases.

06. Choose a method (mixed) *Review: Unit 7, Day 3; 12 course days later.*
Field becomes 1/16; assumes an isolated point charge or an exterior spherically symmetric source.

07. Exit check (exit)
.80×.75=.60 or 60%.

08. STRETCH: Apply it another way (stretch)
Useful 18 W; v=P/(mg)=0.6 m/s; other power 6 W.